

WTI CRUDE OIL FUTURES

Predictive Analysis for Energy Trading Strategies

Instrument Mechanics · Market Participants · Geopolitical Event Study · Predictive Models

Risk Framework

Energy Trading Desks · Portfolio Managers · Corporate Hedgers

01

Instrument Mechanics

How WTI futures work and what each \$1/bbl move means in dollars

02

Market Participants

How hedgers, speculators, and banks position — and why divergence matters

03

Geopolitical Event Study

5 crises, 3 energy eras: does US dominance compress crisis spikes?

04

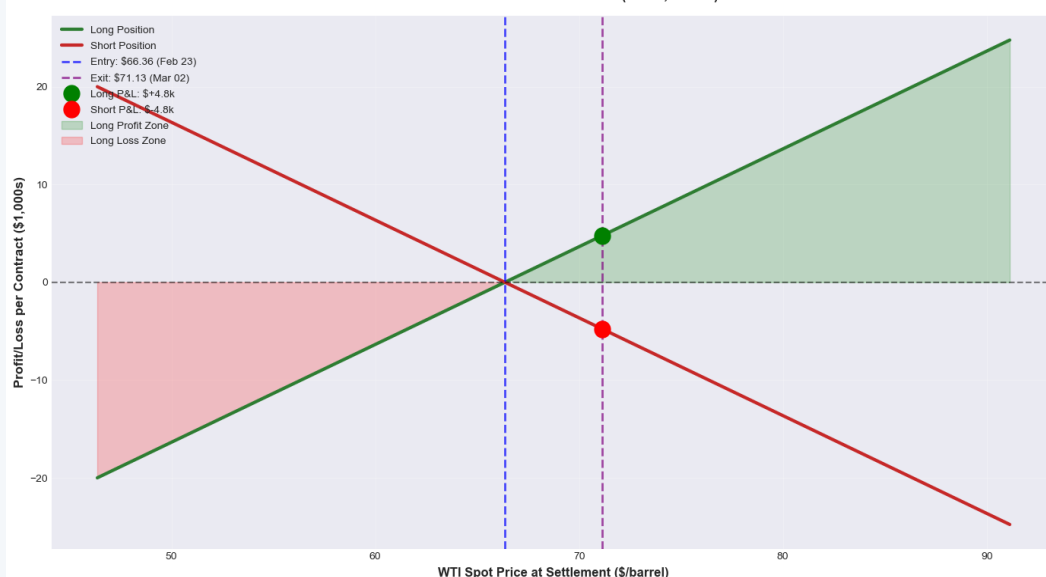
SQL Evidence + Models

12 SQL queries validate the narrative;
2 predictive models add short-horizon signals

Research questions: (1) Can technical indicators predict short-horizon WTI prices? (2) Has US energy independence reduced geopolitical risk premium? (3) Do participant cohorts position differently around crises? (4) Do momentum and volatility signals add predictive value?

One P&L Rule: Every \$1/bbl Move = \$1,000 Per Contract

WTI Futures Payoff Diagram
REAL Market Data: 66.36→71.13 (+4.77, +7.2%)



Contract Specs

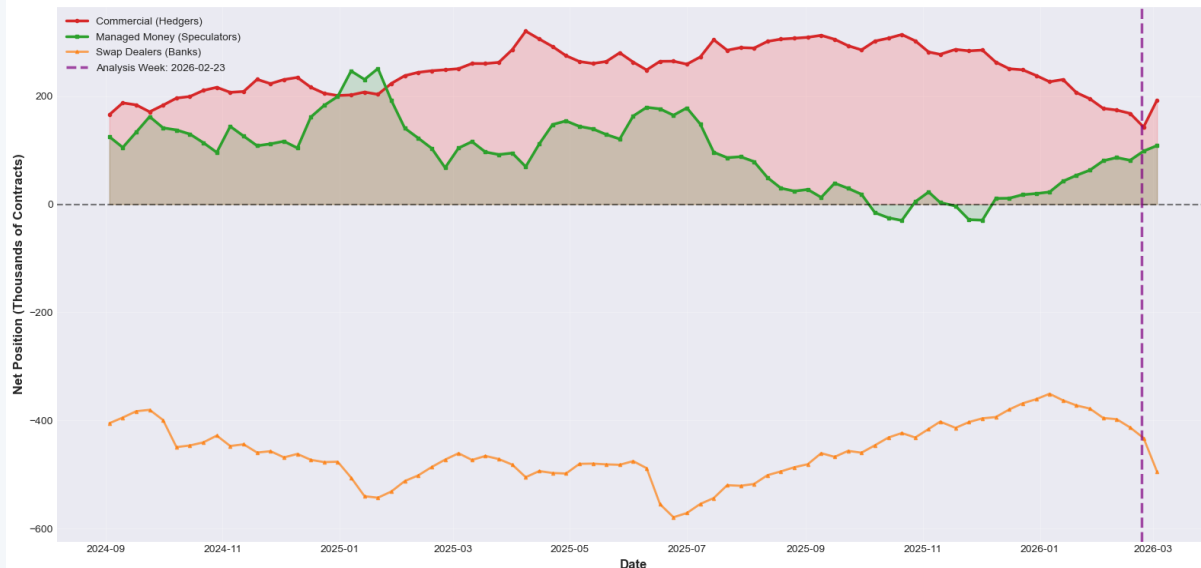
Ticker	CL (NYMEX)
Size	1,000 bbl/contract
Delivery	Cushing, Oklahoma
Tick	\$0.01/bbl = \$10/contract
Last week	\$66.36 → \$71.13 (+\$4.77)
Long P&L	+\$4,800 per contract

FINDINGS

Zero-sum structure means position sizing starts here: $\text{contracts} = \text{risk budget} \div \$1,000\text{-per-}\$1 \text{ move}$.
Hedge ratio, stop distance, and take-profit levels all flow from this one translation.

Extreme Speculator-Hedger Divergence Signals Crowding — and Potential Reversal

WTI Futures: LATEST Market Positioning (18 Months)
From Real CFTC Data - Updated 2026-03-03



Latest Week P&L

(real positioning × real price move)

Commercials

Long 168k net

+\$801M

Managed Money

Long 81k net

+\$387M

Swap Dealers

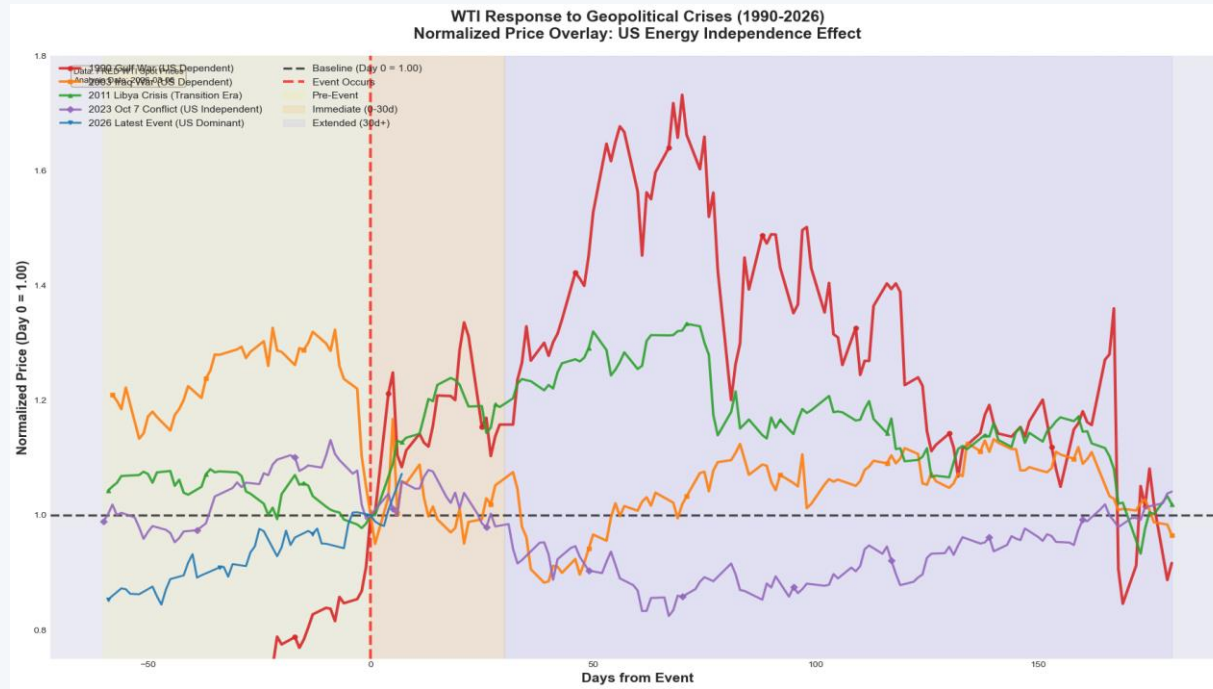
Short 413k net

-\$1.97B

FINDINGS

As of Mar 3 2026: speculators added length (108k net) while commercials also expanded (192k net). When both sides move together, crowding risk is lower — but Swap Dealers' record short (-543k) warrants monitoring. Treat extreme divergence as a regime flag before sizing.

US Energy Dominance Has Structurally Compressed the Geopolitical Risk Premium



US DEPENDENT (1990–2004)

Spikes: 17–34%

TRANSITION (2005–2013)

Spikes: 25%

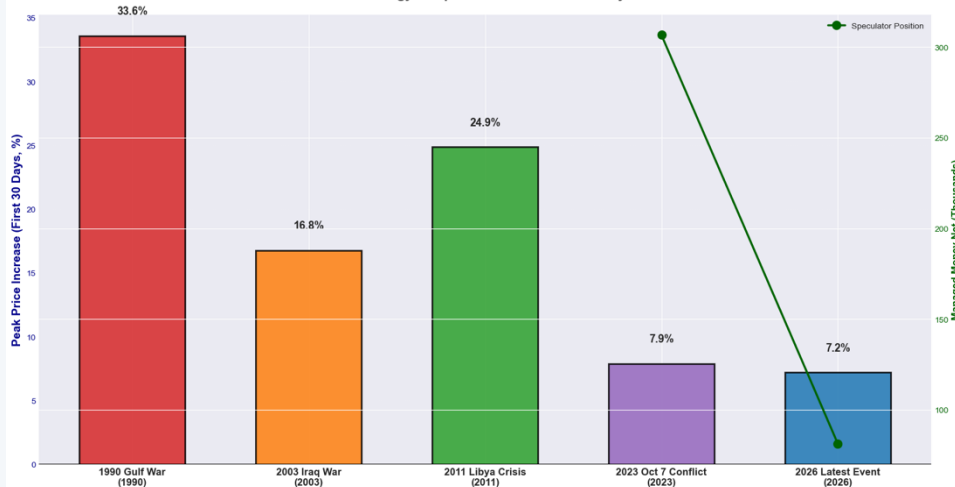
US DOMINANT (2019–present)

Spikes: 7–8%

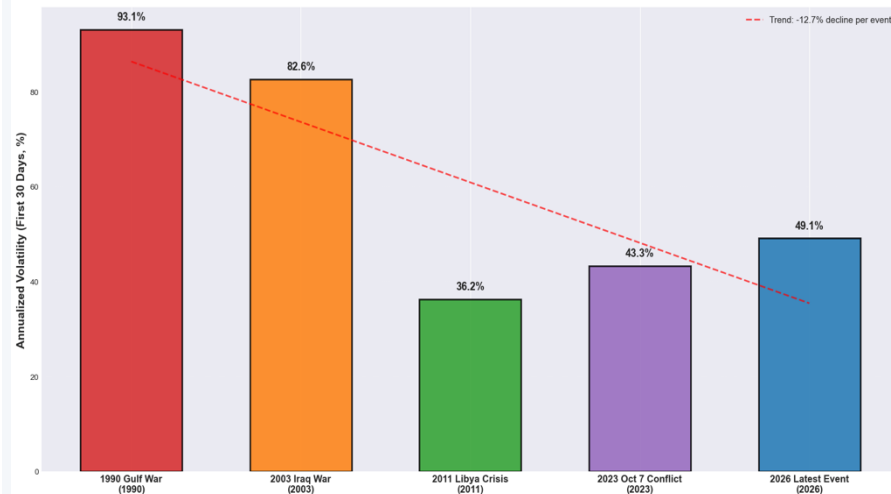
Normalized to Day 0 = 1.00 at each event date. Older crises show much larger and more persistent spikes than recent events.

Peak Response Down 5× and Volatility Down 47% Since the US Dependent Era

Peak WTI Price Response: Evolution Across Decades
with Market Positioning Context
US Energy Independence Reduces Volatility



WTI Price Volatility Following Geopolitical Crises
Market Stability Increases with US Energy Dominance



34% → 7%

Peak 30-day response
1990 vs 2023/2026

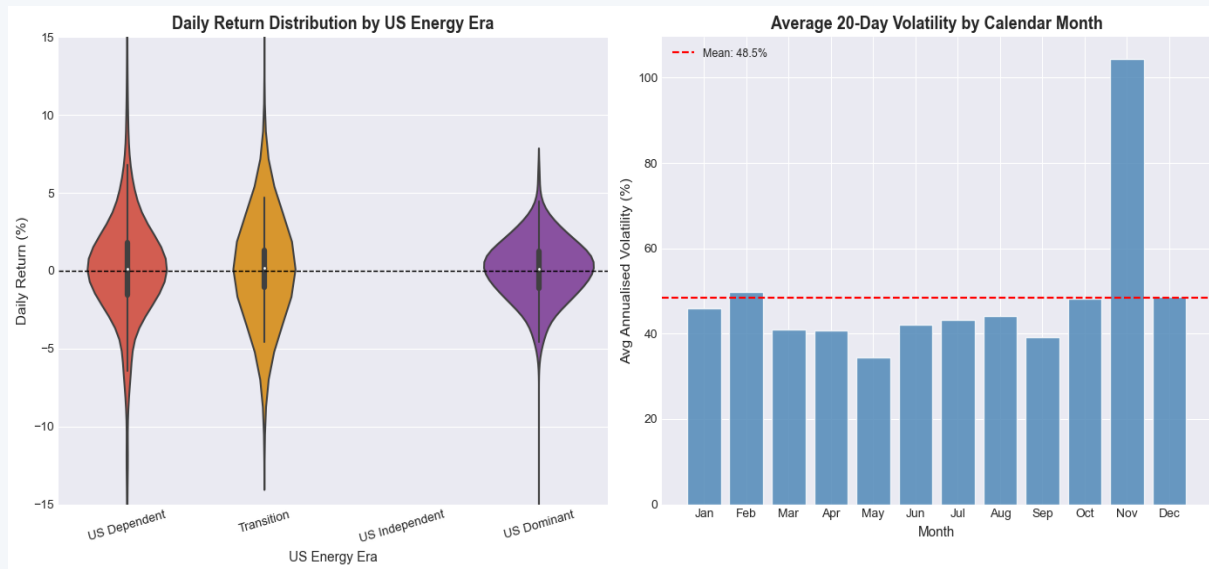
93% → 49%

Annualised volatility
post-event window

No CFTC
→ 306k net long
Speculator position
at Oct 7 (crowded)

Implication: apply era-conditioned stress scenarios — 1990-style VaR shocks are no longer appropriate for geopolitical risk in the US Dominant era.

SQL and EDA Confirm: Era and Month Are the Two Key Regime Dimensions



SQLite Analysis

12 Queries Across 3 Tables

GROUP BY Annual & monthly seasonality

Window fns 30-day rolling avg, price rank

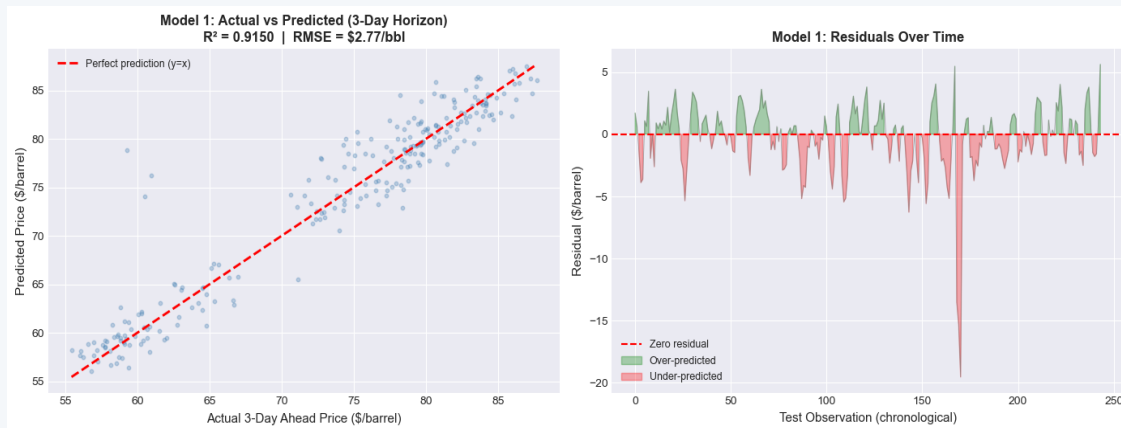
JOINS Event metrics + market context

Subqueries Extreme-vol filtering by era

Key SQL Findings

- US Dominant era: 56% of days in Normal vol regime vs 31% for US Dependent — structurally calmer
- Near-event days (+30d): avg daily return 0.31% vs 0.19% on normal days — mild event premium persists
- Rolling deviation from 30d avg: current WTI sits +7.4 above its 30-day average — elevated but not extreme
- November historically 104% avg volatility — highest seasonal risk month; position size accordingly

Technical Signals Add Modest but Real Predictive Value Beyond Pure Autocorrelation



Model 1: Linear Regression

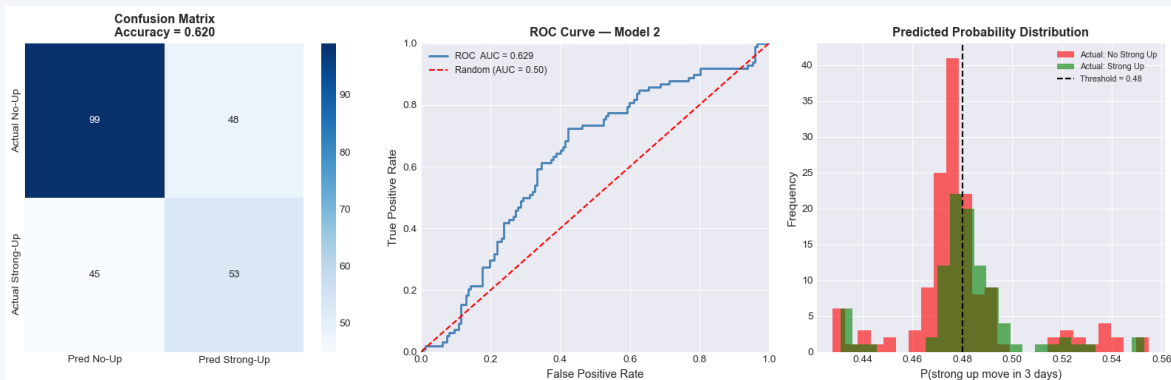
Predict WTI price level 3 trading days ahead

80/20 temporal split — no look-ahead

R^2 **0.915**
91.5% of price variance explained

MAE **\$1.83/bbl**
vs. naive baseline of \$8.26

RMSE **\$2.77/bbl**
≈ 4% of current WTI level



Model 2: Logistic Regression

Predict 3-day strong up-move (>+0.8%)

Threshold = 0.48; avoids look-ahead bias

ROC-AUC **0.629**
vs 0.50 random baseline

Accuracy **62.0%**
vs 60.0% naive baseline

Precision **52.5% | Recall 54.1%**

What Every WTI Practitioner Should Act On

1 Re-calibrate geopolitical stress scenarios to the US Dominant era

Peak crisis response has fallen from ~34% (1990s) to ~7–8% (2023–2026). Applying 1990-era VaR shocks to today's book overstates geopolitical risk premium — freeing capital for other uses

2 Use CFTC positioning divergence as a regime flag, not a standalone signal

Speculator-hedger divergence identifies crowding risk. Combine with event timing and volatility regime before adjusting position size or hedge urgency

3 Layer technical signals into sizing decisions — not direction calls alone

Model 1 ($R^2 = 0.92$, MAE = \$1.83) anchors a 3-day price range for entries. Model 2 (AUC = 0.63) tiers directional conviction. Neither is a standalone buy/sell trigger